Ex 1.4.1

Prove Proposition 1.16: The complex exponential function is periodic with period $2\pi i$. That is, $e^{z+2\pi i} = e^z$ for all z. Furthermore for all $z, w \in \mathbb{C}$ we have

$$e^{z+w} = e^z e^w, \qquad \frac{e^z}{e^w} = e^{z-w}, \qquad e^{-z} = \frac{1}{e^z}.$$

We have $e^z = e^w$ if and only if $w = z + 2\pi i k$ for some $k \in \mathbb{Z}$. In particular, $e^z = 1$ if and only if z is an integer multiple of $2\pi i$. Moreover, $(e^z)^n = e^{nz}$ for all $n \in \mathbb{Z}$.

Pf. From Definition 1.14, we have:

$$e^{x+iy} = e^x \cos y + i e^x \sin y = e^x(\cos y + i \sin y).$$

We show $e^{z+w} = e^z e^w$. Let $z = x + iy$ and $w = u + iv$ (where u, v, x, y $\in \mathbb{Z}$). Then

$$\begin{aligned} e^z e^w &= \big[e^x(\cos y + i \sin y)\big]\big[e^u(\cos v + i \sin v)\big] \\ &= e^{x+u}\big[(\cos y \cos v - \sin y \sin v) + i(\sin y \cos v + \cos y \sin v)\big]. \end{aligned}$$

By the trignometric sum formulas, we have $\cos(y+v)$ and $\sin(y+v)$. We can write,

$$e^z e^w = e^{x+u}\big[\cos(y+v) + i \sin(y+v)\big]$$

Since $z + w = (x+u) + i(y+v)$, we get from Definition 1.14,

$$e^{z+w} = e^{x+u}\big[\cos(y+v) + i \sin(y+v)\big]$$

which matches above. Hence $e^{z+w} = e^z e^w$. □

We show periodicity using the product formula,

$$e^{z+2\pi i} = e^z \cdot e^{2\pi i}.$$

Since $e^{2\pi i} = e^0(\cos 2\pi + i \sin 2\pi) = 1(1+0) = 1$, so $e^{z+2\pi i} = e^z$. □

Further, since $e^z = e^{(z-w)+w} = e^{z-w}e^w$, dividing both sides by e^w (which is nonzero because $|e^w| = e^{\mathrm{Re}(w)} > 0$) gives

$$\frac{e^z}{e^w} = e^{z-w}.$$

We get the quotient in the special case where $z = 0$, which gives $e^{-w} = e^0/e^w = 1/e^w$. □

Let $e^z = e^w$, then we get $e^{z-w} = 1$. Write $z - w = a + ib$ (where $a, b \in \mathbb{R}$). Then

$$|e^{z-w}| = |e^a(\cos b + i \sin b)| = e^a \underbrace{\sqrt{\cos^2 b + \sin^2 b}}_{=1} = e^a.$$

So $e^{z-w} = 1$ causes $e^a = 1$, hence $a = 0$. This leaves $e^{ib} = \cos b + i \sin b = 1$, which requires $\cos b = 1$ and $\sin b = 0$. Both hold simultaneously iff $b = 2\pi k$ for some $k \in \mathbb{Z}$. Therefore $z - w = 2\pi i k$.
In particular, $e^z = 1 = e^0$ iff $z = 2\pi i k$ for some $k \in \mathbb{Z}$. □

For $(e^z)^n = e^{nz}$, where $n \in \mathbb{Z}$. We can check:

- Case $n \geqslant 0$: Use induction on n. The base case $n = 0$ gives $(e^z)^0 = 1 = e^0$. If $(e^z)^n = e^{nz}$, then $(e^z)^{n+1} = e^{nz}e^z = e^{(n+1)z}$ by the product formula.
- Case $n < 0$: Let $n = -m$ (where $m \in \mathbb{Z}^+$). Then

$$(e^z)^n = \frac{1}{(e^z)^m} = \frac{1}{e^{mz}} = e^{-mz} = e^{nz}.$$

Thus shown $(e^z)^n = e^{nz}$ for integer powers. □

Ex 1.4.8(a)

Use (1.6), De Moivre's Formula, and the geometric sum $1 + z + \cdots + z^n = \dfrac{z^{n+1} - 1}{z - 1}$ to prove the following assuming $0 < \theta < 2\pi$:

$$1 + \cos\theta + \cdots + \cos(n\theta) = \frac{1}{2} + \frac{\sin\left((n + \frac{1}{2})\theta\right)}{2\sin(\theta/2)}.$$

Pf. Let $z = e^{i\theta}$. Since $0 < \theta < 2\pi$, we have $z \neq 1$. By De Moivre: $z^k = e^{ik\theta} = \cos(k\theta) + i\sin(k\theta)$. Therefore

$$\sum_{k=0}^{n} \cos(k\theta) = \mathrm{Re}\left(\sum_{k=0}^{n} e^{ik\theta}\right).$$

Since $z = e^{i\theta} \neq 1$,

$$\sum_{k=0}^{n} z^k = \frac{z^{n+1} - 1}{z - 1} = \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1}.$$

Multiply top and bottom by $e^{-i\theta/2}$:

$$\frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1} = \frac{e^{-i\theta/2}\left(e^{i(n+1)\theta} - 1\right)}{e^{-i\theta/2}\left(e^{i\theta} - 1\right)} = \frac{e^{i(n+\frac{1}{2})\theta} - e^{-i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}}.$$

From (1.6), we have $e^{i\alpha} - e^{-i\alpha} = 2i\sin\alpha$, so the denominator is $2i\sin(\theta/2)$. Thus

$$\sum_{k=0}^{n} e^{ik\theta} = \frac{e^{i(n+\frac{1}{2})\theta} - e^{-i\theta/2}}{2i\sin(\theta/2)}.$$

For any complex number α, we have $\mathrm{Re}(\alpha/(2i\sin(\theta/2))) = \mathrm{Im}(\alpha)/(2\sin(\theta/2))$, since dividing by i rotates α by $-\pi/2$, which swaps imaginary and real parts. Then:

$$\begin{aligned}
\mathrm{Re}\left(\frac{e^{i(n+\frac{1}{2})\theta} - e^{-i\theta/2}}{2i\sin(\theta/2)}\right) &= \frac{\mathrm{Im}\left(e^{i(n+\frac{1}{2})\theta}\right) - \mathrm{Im}\left(e^{-i\theta/2}\right)}{2\sin(\theta/2)} \\
&= \frac{\sin\left((n + \frac{1}{2})\theta\right) - (-\sin(\theta/2))}{2\sin(\theta/2)} \\
&= \frac{\sin\left((n + \frac{1}{2})\theta\right) + \sin(\theta/2)}{2\sin(\theta/2)}.
\end{aligned}$$

Split the fraction:

$$\sum_{k=0}^{n} \cos(k\theta) = \frac{\sin(\theta/2)}{2\sin(\theta/2)} + \frac{\sin\left((n + \frac{1}{2})\theta\right)}{2\sin(\theta/2)} = \frac{1}{2} + \frac{\sin\left((n + \frac{1}{2})\theta\right)}{2\sin(\theta/2)}.$$

□

Ex 1.4.9

Observe that the nth-roots of 1 are given by $\{1, \omega, \ldots, \omega^{n-1}\}$ where $\omega = e^{2\pi i/n}$. Assuming $n > 1$, show that $1 + \omega + \cdots + \omega^{n-1} = 0$. Prove a similar result for the nth-roots of -1.

Pf. From Proposition 1.19, the nth-roots of 1 are ω^k for $k = 0, 1, \ldots, n-1$, where $\omega = e^{2\pi i/n}$. Since $n > 1$, we have $\omega \neq 1$, so the geometric sum formula applies:

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = \frac{\omega^n - 1}{\omega - 1}.$$

But $\omega^n = (e^{2\pi i/n})^n = e^{2\pi i} = 1$ by Proposition 1.16, so

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = \frac{1-1}{\omega - 1} = 0.$$

This makes sense geometrically since the nth-roots of unity give vertices on the regular n-gon on the the unit circle, centered at the origin. Symmetrically, the cetntroid should be 0.

Let $-1 = e^{i\pi}$. By Proposition 1.19, the nth-roots of -1 are

$$\zeta_k = e^{i(\pi + 2\pi k)/n} \text{ (for } k = 0, 1, \ldots, n-1)$$

Let $\alpha = \zeta_0 = e^{i\pi/n}$. Then

$$\zeta_k = e^{i\pi/n} e^{2\pi i k/n} = \alpha \omega^k.$$

Factoring out α:

$$\sum_{k=0}^{n-1} \zeta_k = \alpha \sum_{k=0}^{n-1} \omega^k = \alpha \cdot 0 = 0,$$

using the result above. □

Ex 1.4.13

Suppose that $p(z) = a_n z^n + \cdots + a_1 z + a_0$ has real coefficients.

(a) Show that if z_0 is a root of p, then either z_0 is real or $\overline{z_0}$ is also a root of p.

(b) Assuming the Fundamental Theorem of Algebra, show that p must factor as

$$p(z) = a_n(z - r_1)\cdots(z - r_k)\, q_1(z)\cdots q_s(z)$$

where all the r_j are real and each $q_l(z)$ is an irreducible quadratic polynomial with real coefficients and no real roots. (Hint: $(z-w)(z-\overline{w}) = z^2 - 2\,\mathrm{Re}(w)z + |w|^2$.)

(c) Factor $p(z) = z^4 + z^3 + z^2 + z + 1$ into a product of irreducible quadratics. (See Ex 1.4.10.)

Pf. **(a)** We know complex conjugation respects addition and multiplication: $\overline{\alpha+\beta} = \overline{\alpha} + \overline{\beta}$ and $\overline{\alpha\beta} = (\overline{\alpha})(\overline{\beta})$. We do induction on k for "$\overline{z^k} = \overline{z}^k$" ($\forall k \in \mathbb{Z} \geqslant 0$). Since each coefficient a_k is real, $\overline{a_k} = a_k$. Therefore

$$\overline{p(z_0)} = \overline{\sum_{k=0}^{n} a_k z_0^k} = \sum_{k=0}^{n} \overline{a_k} \cdot \overline{z_0}^k = \sum_{k=0}^{n} a_k\, \overline{z_0}^k = p(\overline{z_0}).$$

In the base case, if $p(z_0) = 0$, then $p(\overline{z_0}) = \overline{p(z_0)} = \overline{0} = 0$. Thus, if $z_0 \notin \mathbb{R}$, then $\overline{z_0} \neq z_0$ is a distinct root. □

(b) By the Fundamental Theorem of Algebra, p factors completely over $\mathbb{C}$ as

$$p(z) = a_n(z - w_1)(z - w_2)\cdots(z - w_n).$$

Using part (a), the roots are real ($r_1, \ldots, r_k$) and conjugate pairs ($\{w, \overline{w}\}$ where $\mathrm{Im}(w) \neq 0$). We can group each conjugate pair into the quadratic

$$(z - w)(z - \overline{w}) = z^2 - (w + \overline{w})z + w\overline{w} = z^2 - 2\,\mathrm{Re}(w)\, z + |w|^2.$$

This quadratic has real coefficients. Its discriminant is

$$\Delta = 4\,\mathrm{Re}(w)^2 - 4|w|^2 = 4\big(\mathrm{Re}(w)^2 - \mathrm{Re}(w)^2 - \mathrm{Im}(w)^2\big) = -4\,\mathrm{Im}(w)^2 < 0,$$

so it has no real roots and is therefore irreducible over $\mathbb{R}$. □

(c) Since $z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$, so $p(z) = z^4 + z^3 + z^2 + z + 1$ has roots equal to the primitive 5th-roots of unity $\omega^k = e^{2\pi i k/5}$ for $k = 1, 2, 3, 4$. Since all four roots are non-real, they pair as conjugates: $\overline{\omega} = \omega^4$ and $\overline{\omega^2} = \omega^3$. Therefore, using $\omega + \omega^4 = 2\,\mathrm{Re}(\omega) = 2\cos(2\pi/5)$ and $|\omega|^2 = 1$,

$$(z - \omega)(z - \omega^4) = z^2 - (\omega + \omega^4)z + \omega^5 = z^2 - 2\cos(2\pi/5)\, z + 1,$$
$$(z - \omega^2)(z - \omega^3) = z^2 - (\omega^2 + \omega^3)z + \omega^5 = z^2 - 2\cos(4\pi/5)\, z + 1.$$

By calculation, $\cos(2\pi/5) = (\sqrt{5} - 1)/4$. Since $\cos(4\pi/5) = 2\cos^2(2\pi/5) - 1$, we get $\cos(4\pi/5) = -(\sqrt{5} + 1)/4$. Hence

$$z^4 + z^3 + z^2 + z + 1 = \left(z^2 - \frac{\sqrt{5} - 1}{2}\, z + 1\right)\left(z^2 + \frac{\sqrt{5} + 1}{2}\, z + 1\right).$$

is the product of irreducible quadratics. □

Ex 1.5.2

Let $c \in \mathbb{C} \setminus \{0\}$ and $a \in \mathbb{C}$. Show that the geometric series

$$\sum_{n=0}^{\infty} ca^n$$

converges to $\frac{c}{1-a}$ if $|a| < 1$, and diverges otherwise.

Pf. For $a = 1$: every partial sum is $s_N = c(N+1)$, so $|s_N| \to \infty$ and the series diverges.

For $a \neq 1$, we use the telescoping trick. Since $(1-a)(1+a+\cdots+a^N) = 1-a^{N+1}$, the partial sum is

$$s_N = c\sum_{n=0}^{N} a^n = c \cdot \frac{1-a^{N+1}}{1-a}.$$

If $|a| < 1$, then $|a^{N+1}| = |a|^{N+1} \to 0$ as $N \to \infty$ (since $|a| < 1$ implies $|a|^k$ decreases, tending to 0). Therefore

$$\lim_{N\to\infty} s_N = c(\frac{1-0}{1-a}) = \frac{c}{1-a}.$$

If $|a| > 1$, then $|a|^{N+1} \to \infty$, so $|a^{N+1}| \to \infty$ and s_N does not converge.
If $|a| = 1$ (including $a = 1$ as above), then $|ca^n| = |c||a|^n = |c| > 0$ for all n, so the terms ca^n do not tend to zero. But since the terms of any convergent series must tend to 0, the series diverges. □

Comparing with Ex 1.5.9: Taking $c = 1$ gives $\sum_{n=0}^{\infty} a^n = 1/(1-a)$ for $|a| < 1$. This is the power series for $1/(1-z)$ centered at 0. Intuitively, the partial sums need more and more terms to get close to $1/(1-z)$ on negative end.

Ex 1.5.8

Prove Dini's Theorem: Suppose that (h_n) is a sequence of continuous nonnegative functions on a compact set K converging monotonically to zero on K. Then $h_n \searrow 0$ uniformly on K.

Pf. Fix $\varepsilon > 0$. For each $x \in K$, the sequence $h_n(x)$ decreases to 0, so there exists an index k such that $h_k(x) < \varepsilon/2$. Since h_k is continuous, there is an open neighborhood U_x of x s.t.

$$h_k(y) < \varepsilon \quad \text{for all } y \in U_x \cap K.$$

(By continuity we are below ε in a neighborhood.)

The collection $\{U_x : x \in K\}$ is an open cover of K. Since K is compact, finitely many of them suffice:

$$K \subseteq U_{x_1} \cup U_{x_2} \cup \cdots \cup U_{x_m}.$$

Define $N = \max(N_{x_1}, N_{x_2}, \ldots, N_{x_m})$.

Let $n \geqslant N$ and let $y \in K$ be arbitrary. Then $y \in U_{x_j}$ for some j. Since the sequence is monotonically decreasing and $n \geqslant N \geqslant N_{x_j}$, we have

$$0 \leqslant h_n(y) \leqslant h_{N_{x_j}}(y) < \varepsilon.$$

This holds for all $y \in K$ simultaneously, so $\sup_{y \in K} h_n(y) \leqslant \varepsilon$ whenever $n \geqslant N$. I.e., $h_n \searrow 0$ uniformly on K. $\square$

Ex 2.1.8

Let $A_r(z_0) = \{z \in \mathbb{C} : |z - z_0| > r > 0\}$ and suppose that

$$g(z) = \sum_{j=1}^{\infty} \frac{b_j}{(z - z_0)^j}$$

converges for all $z \in A_r(z_0)$ as in Ex 1.5.10. Show that both g and g' are holomorphic on $A_r(z_0)$ and that

$$g'(z) = \sum_{j=1}^{\infty} \frac{-j\, b_j}{(z - z_0)^{j+1}} \quad \text{for all } z \in A_r(z_0).$$

Show also that the coefficients b_j appearing above are uniquely determined by g.

Pf. Let $w = 1/(z - z_0)$. When $z \in A_r(z_0)$ (i.e. $|z - z_0| > r$), we get $|w| = 1/|z - z_0| < 1/r$. So as z ranges over $A_r(z_0)$, the variable w ranges over the disk $D_{1/r}(0) \setminus \{0\}$.
In terms of w we get the series as

$$h(w) := g(z_0 + 1/w) = \sum_{j=1}^{\infty} b_j w^j, \qquad 0 < |w| < 1/r.$$

Since $h(0) = 0$, this is actually a power series in w centered at 0 that converges on all of $D_{1/r}(0)$. Indeed, by Ex 1.5.10 the radius of convergence of $\sum b_j w^j$ is at least $1/r$.

By Theorem 2.4, h is holomorphic on $D_{1/r}(0)$ and can be differentiated on each term:

$$h'(w) = \sum_{j=1}^{\infty} j\, b_j\, w^{j-1}, \qquad |w| < 1/r.$$

The function $\varphi(z) = 1/(z - z_0)$ is holomorphic on $A_r(z_0)$ with $\varphi'(z) = -1/(z - z_0)^2$. Since $g(z) = h(\varphi(z))$, the chain rule gives

$$g'(z) = h'(\varphi(z))\varphi'(z) = \left(\sum_{j=1}^{\infty} j\, b_j \frac{1}{(z - z_0)^{j-1}} \right) \frac{-1}{(z - z_0)^2} = \sum_{j=1}^{\infty} \frac{-j\, b_j}{(z - z_0)^{j+1}}.$$

Since h is holomorphic, and φ is holomorphic, the composition $g = h \circ \varphi$ is holomorphic on $A_r(z_0)$. The same argument applied to g' (or equivalently h') shows g' is also holomorphic here. □

By Corollary 2.7, the coefficients of the power series $h(w) = \sum_{j=0}^{\infty} b_j w^j$ are uniquely determined by h: specifically $b_j = h^{(j)}(0)/j!$. Since h is completely determined by g (via $h(w) = g(z_0 + 1/w)$), the coefficients b_j are uniquely determined by g. □

Ex 2.2.3

Suppose that D is a domain and $f \in H(D)$. Show that if f is real-valued in D, then f is constant.

Pf. Let $f(z) = u(z) + iv(z)$ where $u = \mathrm{Re}(f)$ and $v = \mathrm{Im}(f)$ are real-valued. The hypothesis that f is real-valued means $v(z) = 0$ for all $z \in D$. I.e., the partial derivatives become $v_x \equiv 0$ and $v_y \equiv 0$ on D.

Since $f \in H(D)$, by Cauchy-Riemann on every point of D: $u_x = v_y = 0$ and $u_y = -v_x = 0$. So both first partial derivatives of u zero out on D.

Since D is a domain (open and connected) and the continuous function u has $u_x \equiv 0$ and $u_y \equiv 0$ throughout D, Proposition 1.7 (i.e. function with vanishing partials on an open connected set is constant) gives that u is constant on D.
Therefore $f = u + iv = u$ is constant on D. □

Ex 2.2.5

Let $f : D \subset \mathbb{C} \to \mathbb{C}$ be a function of the form $f(z) = u(z) + iv(z)$, and let $F : D \subset \mathbb{R}^2 \to \mathbb{R}^2$ be the companion function $F(x, y) = (u(x, y), v(x, y))$. Let $z_0 = x_0 + iy_0 \in D$. Show that $f'(z_0)$ exists if and only if the Cauchy–Riemann equations hold at z_0 and F is differentiable at (x_0, y_0).

Pf. Suppose the Cauchy-Riemann holds at z_0, s.t. $u_x = v_y$ and $u_y = -v_x$ there. Let $w = u_x(z_0) + iv_x(z_0) = f_x(z_0)$, and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map with matrix

$$[T] = \begin{pmatrix} u_x(z_0) & -v_x(z_0) \\ v_x(z_0) & u_x(z_0) \end{pmatrix} = \begin{pmatrix} u_x(z_0) & u_y(z_0) \\ v_x(z_0) & v_y(z_0) \end{pmatrix},$$

where the second equality uses the Cauchy-Riemann fo the Jacobian of F at (x_0, y_0).

Let $z = h + ik$ with $h, k \in \mathbb{R}$. Using the hint in the question and equation (2.13), we get

$$\|F(x_0 + h,\ y_0 + k) - F(x_0, y_0) - T(h, k)\| = |f(z_0 + z) - f(z_0) - wz|. \tag{$\star$}$$

($\Longrightarrow$) If $f'(z_0)$ exists. As in class and Ex 2.2.1, we get $f'(z_0)$ by restricting h in the difference quotient to real values (which gives $f_x(z_0) = f'(z_0)$), and to only imaginary values (which gives $-if_y(z_0) = f'(z_0)$). Thus, we combine to get $f_x(z_0) = -if_y(z_0)$, which is the Cauchy-Riemann at z_0.
The identity $(\star)$ works with $w = f'(z_0)$. Since $f'(z_0)$ exists, the right side of $(\star)$ satisfies

$$|f(z_0 + z) - f(z_0) - wz| = o(|z|) \quad \text{as } z \to 0.$$

By $(\star)$, $\|F(x_0 + h, y_0 + k) - F(x_0, y_0) - T(h, k)\| = o(\|(h, k)\|)$, which is F differentiable at (x_0, y_0) with derivative T.

($\Longleftarrow$) If Cauchy-Riemann holds and F is differentiable. Cauchy-Riemann ensures the matrix $[T]$ has the form above, and F being differentiable at (x_0, y_0) means the left side of $(\star)$ is $o(\|(h, k)\|)$. By $(\star)$,

$$|f(z_0 + z) - f(z_0) - wz| = o(|z|) \quad \text{as } z \to 0.$$

Dividing by $|z|$ and taking $z \to 0$ gives

$$f'(z_0) = w = u_x(z_0) + iv_x(z_0) = f_x(z_0).$$

Hence shown $f'(z_0)$ exists iff Cauchy-Riemann holds at z_0 and F is differentiable at (x_0, y_0). □